Advanced Programming Practice · 21CSC203P

Unit 4 · Session 2 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 2 : SLO â€“ 2 | Built-in Higher-Order Functions (map, reduce, filter)

■ I. Fill in the Blanks

1. The filter() function in Python returns an object that contains elements from the iterable that satisfy a condition (predicate function).
2. A function is pure if it does not have side effects and does not change any external (global) state.

■ II. Multiple Choice

1. list(map(lambda x: x + 1, [1, 2, 3])) â†’ [2, 3, 4] â€” map applies the lambda to each element, adding 1 to each.

■ III. Extra Examples

• filter: list(filter(lambda x: x % 2 == 0, range(10))) â†’ [0, 2, 4, 6, 8].
• reduce: functools.reduce(lambda a, b: a * b, [1, 2, 3, 4]) â†’ 24 (factorial of 4).